

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I M.VAISHNAV (192524251) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M.VAISHNAV (192524251)

Question 1: Backtracking Search – Doctor Shift Assignment

Variables

Doctors: D1, D2, D3

Domains

Morning (M), Afternoon (A), Night (N)

Constraints

1. D1 cannot work Night shift.
2. D2 must work before D3 (Morning < Afternoon < Night).
3. Only one doctor per shift.
4. D3 cannot work Morning.
5. All doctors must be assigned exactly one shift.

Step 1: Assign D1

Possible shifts:

- Morning ✓
- Afternoon ✓
- Night ✗

Choose:

D1 = Morning

Current Assignment:

D1 → Morning

Step 2: Assign D2

Remaining shifts:

- Afternoon
- Night

Choose:

D2 = Afternoon

Current Assignment:

D1 → Morning

D2 → Afternoon

Step 3: Assign D3

Remaining shift:

Night

Check Constraints:

- D3 $\neq$ Morning ✓
- D2 before D3 (Afternoon < Night) ✓
- One doctor per shift ✓
- All doctors assigned ✓

Final Valid Schedule

Shift	Doctor
Morning	D1
Afternoon	D2
Night	D3

Conclusion

The backtracking search successfully finds a valid assignment:

D1 → Morning, D2 → Afternoon, D3 → Night

Question 2: BFS Search Algorithm

Given Grid (5×5)

S = Start, G = Goal, X = Obstacle

S 0 0 X 0

0 X 0 X 0

0 X 0 0 0

0 0 X X 0

0 0 0 0 G

Constraints

1. Movement allowed: Up, Down, Left, Right
2. Cost per move = 1
3. Obstacles cannot be crossed
4. Manhattan Distance Heuristic

Manhattan Distance Formula

$$h(n) = |x_g - x_n| + |y_g - y_n|$$

Goal Position = (4,4)

BFS Node Expansion

Initial Queue:

[(0,0)]

Visited:

{(0,0)}

Expand (0,0)

Neighbors:

(1,0), (0,1)

Queue:

[(1,0), (0,1)]

Expand (1,0)

Neighbor:
(2,0)

Queue:
[(0,1), (2,0)]

Expand (0,1)

Neighbor:
(0,2)

Queue:
[(2,0), (0,2)]

Expand (2,0)

Neighbor:
(3,0)

Queue:
[(0,2), (3,0)]

Expand (0,2)

Neighbor:
(1,2)

Queue:
[(3,0), (1,2)]

Expand (3,0)

Neighbors:
(4,0), (3,1)

Queue:
[(1,2), (4,0), (3,1)]

Expand (1,2)

Neighbor:
(2,2)

Queue:

[(4,0), (3,1), (2,2)]

Continue Expansion

(4,0)

(3,1)

(2,2)

(4,1)

(2,3)

(4,2)

(2,4)

(4,3)

(3,4)

(4,4) Goal Reached

Optimal Path

(0,0)

→ (0,1)

→ (0,2)

→ (1,2)

→ (2,2)

→ (2,3)

→ (2,4)

→ (3,4)

→ (4,4)

Cost Calculation

Number of Moves = 8

Cost per Move = 1

Total Cost = $8 \times 1 = 8$

Final Answer

Optimal Path:

S → (0,1) → (0,2) → (1,2) → (2,2) → (2,3) → (2,4) → (3,4) → G

Optimal Cost = 8

Question 3

(a) Constraint Analysis

1. Movement Constraint

The robot can move only Up, Down, Left, and Right. Diagonal movement is not allowed. This limits the available actions and affects path planning.

2. Obstacles

Blocked cells cannot be traversed. The robot must identify alternative routes whenever obstacles are encountered.

3. Partial Observability

The robot can observe only adjacent cells. It does not have complete knowledge of the environment and must discover information while moving.

4. Movement Cost

Each movement has a cost of 1. The robot should prefer shorter paths to reduce the total travel cost.

5. Risky Zones

Entering a risky zone adds an additional cost of 2. The robot should avoid risky areas whenever possible to minimize overall cost.

6. Dynamic Knowledge Update

The robot continuously gathers information about the environment. It must update its knowledge base and revise decisions when new obstacles or risks are detected.

Effect on Decision-Making

These constraints require the robot to balance safety, cost, and exploration. The robot must continuously evaluate available actions and choose the safest and least expensive route.

(b) Solution Approach Using Uniform Cost Search and Online Search Agent

Step 1

The robot starts at position S.

Step 2

The robot observes all adjacent cells using its sensors.

Step 3

Possible legal moves are generated.

Step 4

The total cost of each move is calculated.

Total Cost = Movement Cost + Risk Cost

Step 5

Uniform Cost Search selects the node with the lowest cumulative cost.

Step 6

The robot moves to the selected node.

Step 7

The robot updates its knowledge base with newly discovered information.

Step 8

The process repeats until the robot reaches the survivor at G.

Example

Path:

$S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$

Costs:

$S \rightarrow A = 1$

$A \rightarrow B = 1$

$B \rightarrow C = 3$ (1 movement + 2 risk)

$$C \rightarrow G = 1$$

$$\text{Total Cost} = 1 + 1 + 3 + 1 = 6$$

The robot compares all possible routes and chooses the route with the minimum total cost.

(c) Application of AI Concepts

Intelligent Agent

The rescue robot acts as an intelligent agent.

Percepts:

- Adjacent cells
- Obstacles
- Risk zones

Actions:

- Move Up
- Move Down
- Move Left
- Move Right

The robot senses its environment and takes actions to achieve its goal.

Rationality

A rational agent selects actions that maximize performance and minimize cost.

Example:

- Safe Path Cost = 5
- Risky Path Cost = 8

The robot chooses the safe path because it provides a better outcome.

Environment Type

1. Partially Observable – Robot can observe only nearby cells.
2. Single Agent – Only one robot is operating.
3. Dynamic Environment – New information becomes available during execution.

4. Discrete Environment – Movement occurs between grid cells.
 5. Deterministic Environment – Actions lead to predictable results.
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(d) Justification of Chosen Approach

The combination of Uniform Cost Search and Online Search Agent is the most suitable approach because:

1. It works effectively in unknown and partially observable environments.
2. It dynamically updates information while exploring.
3. It considers both movement cost and risk cost.
4. It always expands the lowest-cost node first.
5. It guarantees an optimal solution when costs are non-negative.
6. It ensures safe and efficient navigation to the survivor.

Conclusion

The rescue robot operates as a rational intelligent agent in a partially observable environment. By using Uniform Cost Search along with Online Search techniques, it can dynamically discover information, avoid risky zones, minimize total cost, and successfully reach the survivor while satisfying all given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately.	Identifies most constraints with minor gaps.	Partial understanding; misses key constraints.	Fails to identify constraints correctly.
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically.	Provides reasonable analysis with some prioritization.	Limited analysis; weak prioritization.	No meaningful analysis or prioritization.
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints.	Develops workable solutions with minor limitations.	Solutions partially satisfy constraints.	Solutions do not meet constraints.
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools.	Applies concepts with minor errors.	Limited application; requires guidance.	Unable to apply relevant concepts.
Justification	10%	Provides strong justification for decisions with logical reasoning.	Provides justification with some clarity.	Weak or unclear justification.	No justification provided.